

Task 5: Exploratory Data Analysis (EDA) - Titanic Dataset

Objective

To explore the Titanic dataset using statistical summaries and visualizations in order to identify patterns, trends, relationships, and anomalies.

Dataset Overview

The Titanic dataset contains information about 891 passengers and 12 features including PassengerId, Survived, Pclass, Name, Sex, Age, Fare, SibSp, Parch, Ticket, Cabin, and Embarked.

Data Exploration

The dataset was explored using `df.info()`, `df.describe()`, `isnull().sum()`, and `value_counts()`. Age contained missing values, while Cabin had a large number of missing entries. Passenger class and gender distributions were also examined.

Visualizations and Observations

1. Age Distribution Histogram

Most passengers were between 20 and 40 years old.

2. Fare Distribution Boxplot

The Fare feature contains several outliers, indicating that some passengers paid significantly higher fares.

3. Survival by Gender

Female passengers had a noticeably higher survival rate compared to male passengers.

4. Survival by Passenger Class

Passengers traveling in first class had better survival chances than those in second and third class.

5. Scatter Plot (Age vs Fare)

Passengers paying higher fares generally showed better survival outcomes.

6. Correlation Heatmap

Survival showed a positive relationship with Fare and a negative relationship with Passenger Class (Pclass).

7. Pairplot

The pairplot helped visualize relationships among Age, Fare, Pclass, and Survival.

Summary of Findings

1. Female passengers had higher survival rates.
2. First-class passengers survived more often.
3. Fare contains several outliers.
4. Most passengers were between 20 and 40 years old.
5. Higher ticket fares were associated with better survival chances.
6. Pclass and Fare significantly influenced survival.

Conclusion

The exploratory analysis revealed that gender, passenger class, and fare were important factors influencing survival. Visualizations and statistical summaries helped uncover meaningful patterns and relationships within the Titanic dataset.